From No Clock to Relational Time:
A Page–Wooters Genesis of Clocks, Multiplicities, and Geometry

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Abstract

We present a self-contained, experimentally checkable construction of *time as a relational observable*. A global, clockless constraint a la Page–Wooters (PW) is imposed on a composite Hilbert space $(S \otimes C)$. "Dynamics" appears only after conditioning on the clock C, which we realize as a binary spin-structure regulator (PBC vs APBC) on a finite cycle. From the conditioned viewpoint we recover: (i) the universal decimated two-step law (AR(2)) and its exact Green functions, (ii) clock "births" from net-zero probes with an envelope that sets spiral pitch and speed limits, (iii) multiplicities of species labeled by operational invariants, and (iv) a finite-loop bridge from envelope holonomies to Levi–Civita transport with explicit error budgets. We supply proofs, minimal experiments, figures, and Python snippets for reproducibility.

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1 Axiom 0: Clockless Universe (PW Constraint)

Global, timeless state

Let $\mathcal{H}_S$ (system) carry a two-component engine and let $\mathcal{H}_C$ (clock) be a finite register. We postulate a joint state $|\Psi\rangle \in \mathcal{H}_S \otimes \mathcal{H}_C$ that satisfies the *timeless* constraint

$$(M_S \otimes M_C)\,|\Psi\rangle \;=\; |\Psi\rangle, \tag{1}$$

the discrete analogue of $(H_S + H_C)|\Psi\rangle = 0$. There is no background time; the constraint

encodes a relational invariant.

Clock spin structure as regulator. We realize the clock as a one-dimensional register with a $\mathbb{Z}_2$ boundary twist: *periodic* (PBC/Ramond) vs *anti-periodic* (APBC/Neveu–Schwarz). APBC gaps the clock Dirac spectrum; PBC admits a zero mode. Conditioning is well-posed only in the gapped sector; this binary choice is a falsifiable prediction.

2 Relational Dynamics and the Universal AR(2) Law

Conditional evolution (PW)

Fix a readout basis $\{|t\rangle_C\}$ on the clock. The *conditional* system state $|\psi_t\rangle_S \propto {}_C\langle t|\Psi\rangle$ obeys

$$|\psi_{t+1}\rangle_S = M_S\,|\psi_t\rangle_S. \tag{2}$$

Sketch

From (1), $(M_S \otimes M_C)|\Psi\rangle = |\Psi\rangle$. Using $M_C|t\rangle = |t+1\rangle$ and applying ${}_C\langle t+1|$ gives ${}_C\langle t+1|\Psi\rangle = M_S\,{}_C\langle t|\Psi\rangle$, i.e., the stated update.

Cayley–Hamilton ⇒ universal decimated law

Let $M_S \in \mathrm{SL}(2,\mathbb{C})$ with $V_\ell = \mathrm{tr}(M_S^\ell)$. For any linear observable $x_t = w^\dagger|\psi_t\rangle$, define residue tracks $y_t^{(r)} = x_{r+\ell t}$. Then

$$y_{t+2}^{(r)} - V_\ell\, y_{t+1}^{(r)} + y_t^{(r)} = 0, \qquad r = 0,\dots,\ell-1. \tag{3}$$

Explicit 2×2 derivation

Cayley–Hamilton for M_S with $\det M_S = 1$ gives $M_S^{2\ell} - V_\ell M_S^\ell + \mathsf{I} = 0$. Act on $|\psi_t\rangle$ and project by $w^\dagger$ at steps $t, t+\ell, t+2\ell$ to get (3).

Roots, pitch, and speed. Equation (3) has roots $r_\pm = \frac{1}{2}(V_\ell \pm \sqrt{V_\ell^2 - 4})$. Write $r_+ = \rho e^{i\theta}$; the *spiral pitch* is $\Pi = \log\rho/\theta$. A graph-local envelope $E = e^{-\sigma^2\kappa\,\mathsf{L}_\alpha^\dagger\mathsf{L}_\alpha}$ suppresses propagation and sets a speed limit $v_{\max} \lesssim a \sup\|\nabla\varepsilon\|/(1+\sigma^2\kappa\Lambda_{\max}^2)$.

3 From No Clock to a Clock: Births via Net-Zero Probes

Net-zero probe and exact Green function. For $Q = 1$ and $V = 2\cos\phi$, the causal Green function is $G_n = \sin(n\phi)/\sin\phi$. On a macrocycle of length L choose a net-zero source $\sum_{m\,(\mathrm{mod}\,L)} s_m = 0$ (e.g. $\delta_{m,0} - \delta_{m,L/2}$). The relational response is $x_m = \sum_{n\le m} G_{m-n}s_n$. An envelope E localizes the birth and fixes the spiral pitch.

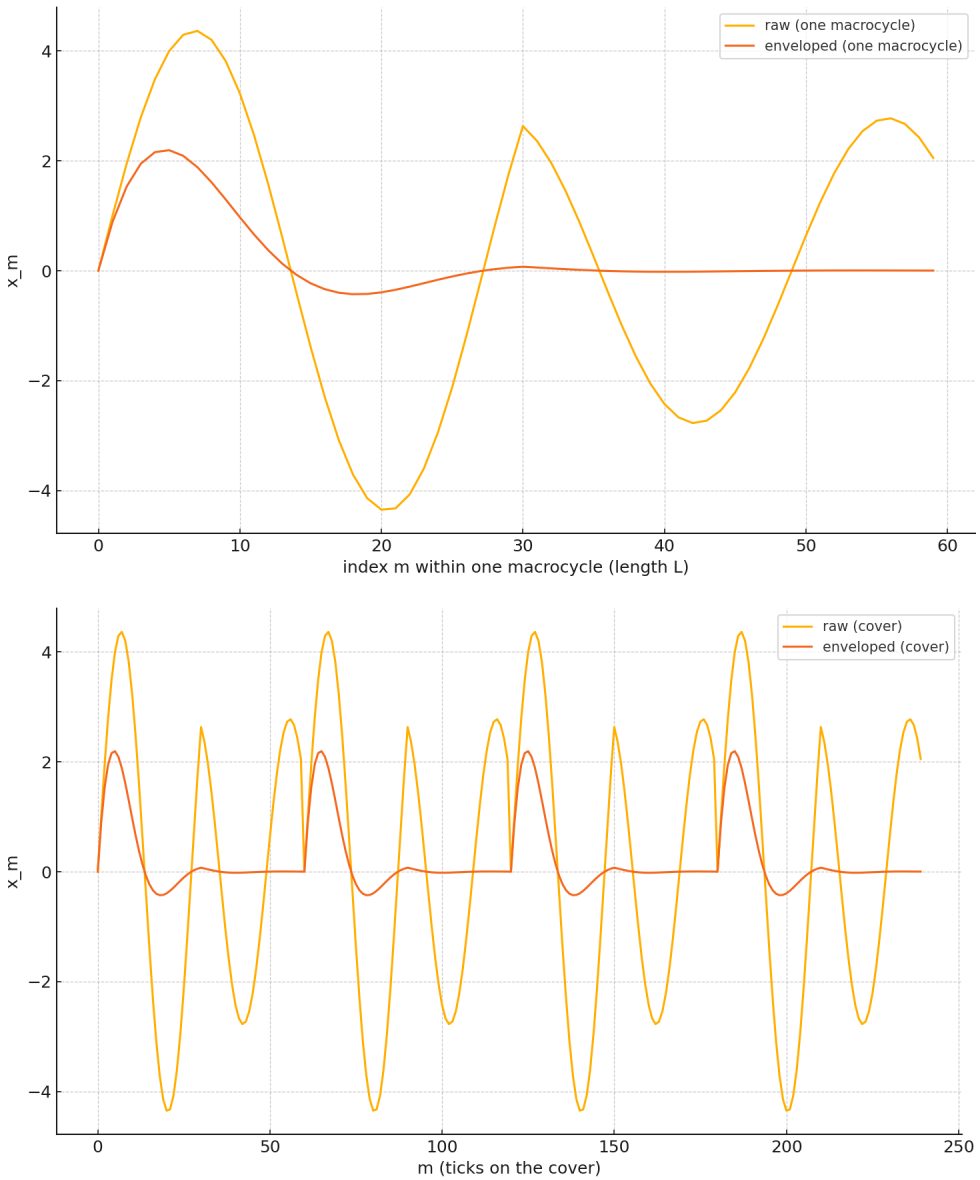

Figure 1: **Clock births from net-zero probes.** Top: one macrocycle (raw vs enveloped). Bottom: repeated on the universal cover (4 periods).

4 Binary Spin Structure: PBC Zero Mode vs APBC Gap

Dirac on a time circle (discrete). Using the lattice derivative, the clock spectrum is $E(k) = \frac{2}{\Delta\tau}\sin k$ with $k = \frac{2\pi}{N}(m + \frac{s}{2})$ and $s \in \{0, 1\}$ (PBC vs APBC). Thus PBC admits a zero mode; APBC has a gap $\gtrsim \frac{2}{\Delta\tau}\sin(\pi/N)$.

Binary holonomy prediction (falsifiable). If one macrocycle contains L steps, APBC shifts the phase by π/L, so $V_L^{\mathrm{APBC}} = -V_L^{\mathrm{PBC}}$ while Q^L is unchanged.

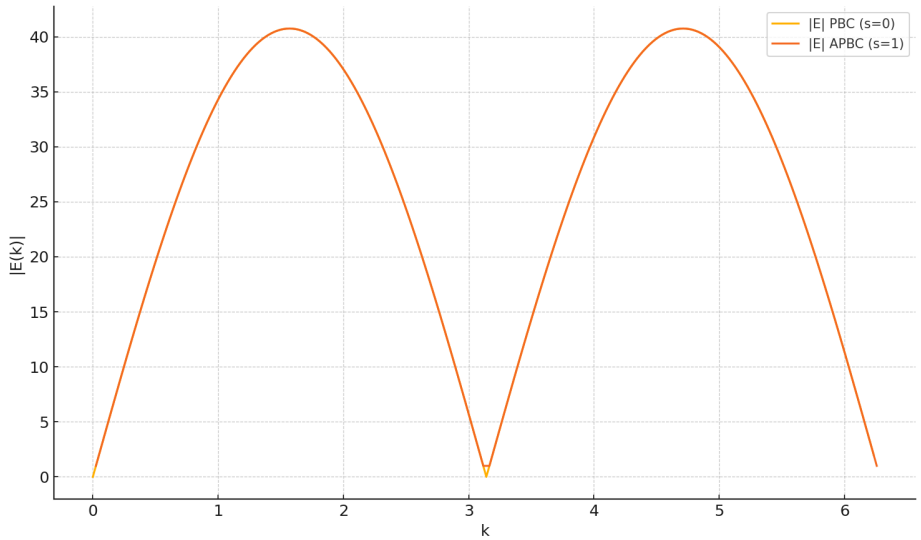

Figure 2: **Discrete Dirac on a time circle.** PBC shows a zero mode; APBC is gapped.

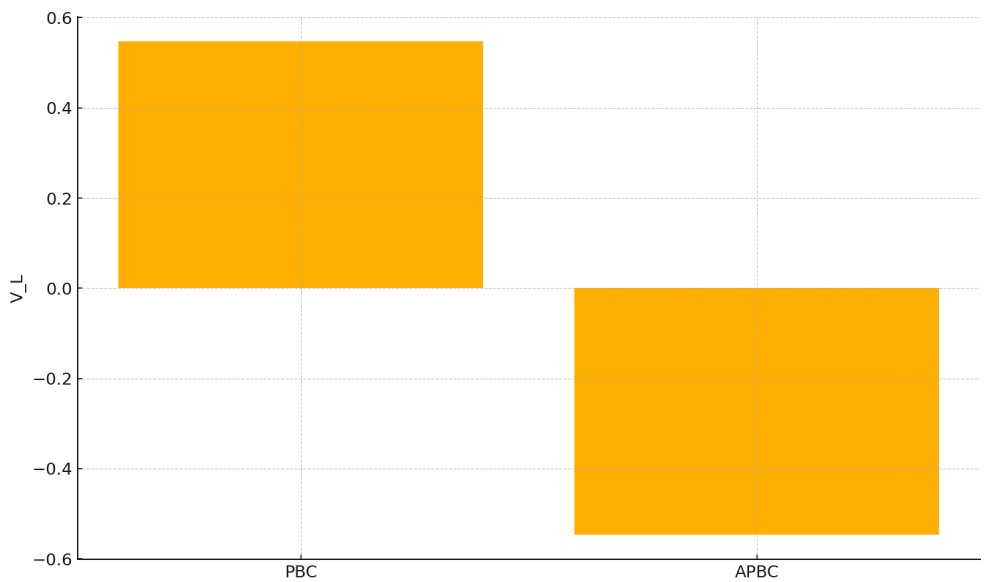

Figure 3: **Binary holonomy test.** V_L flips sign under APBC, as predicted.

5 Multiplicities and Species (Operational Invariants)

We label species by $\Lambda = (\ell, V_\ell, \rho, \theta, \chi, (c_0, c_1), \mathrm{hol})$. Checksums (c_0, c_1) classify bands/gaps (Hofstadter dictionary); χ encodes the PBC/APBC parity; holonomy distinguishes circle vs Möbius ($\mathrm{Pin}^{\pm}$).

Independence scale and counts

If the envelope has gap $\mu > 0$, correlations between residue tracks separated by d decay as $\sim e^{-\mu d}$. Tracks beyond $d > \ell$ are effectively independent; counts per label Λ yield multiplicities.

Strong non-commutativity. For composites $\{M_j\}$, the Magnus/BCH expansion $\log M_{\text{eff}} = \sum_j \log M_j + \frac{1}{2}\sum_{i<j}[\log M_i, \log M_j] + \cdots$ quantifies trace splitting and hence label bifurcations.

6 Emergent Space from Conditioned Phases; GR Bridge

Coframes from clocks. Let $e^a = \Theta^a{}_i\, dx^i$ be the pushforward of conditioned phases. Parallel transport under the envelope approximates Levi–Civita holonomy with an explicit finite-loop bound.

Finite-loop bridge (error budget)

For a loop C bounding $\Sigma(C)$,

$$\|P_{\text{env}}(C) - H_{\text{LC}}(C)\| \;\le\; C\Big(\sqrt{\textstyle\int_{\Sigma(C)}\|F\|^2\, dA} \;+\; h \;+\; \ell^3\|\nabla\Omega\|_\infty \;+\; \mu^{-1}\sigma^2\kappa\,\ell\,\tau\Big), \tag{4}$$

with $F = \partial_i\Theta^a{}_j - \partial_j\Theta^a{}_i$ and mesh h.

Idea of proof

(1) Strang split one edge as $U_e = S^{1/2}TS^{1/2}$ and use BCH to track commutators; (2) non-Abelian Stokes to compare surface-ordered exponentials; (3) Cartan + Stokes + Poincaré/trace to bound boundary terms by $\|F\|_{L^2(\Sigma)}$; (4) accumulate leakage from the envelope via its spectral gap.

7 Directed/Weighted Graphs and Scaling/Tuning

For directed or weighted graphs, use $\mathsf{L}_{\alpha\text{psd}} = (\mathsf{I} - \alpha\mathsf{A})^\dagger(\mathsf{I} - \alpha\mathsf{A})$ and replace spectral radii by singular values. Speed and holonomy bounds adapt up to constants. Scaling rules (spacing a, step τ, $\Lambda_{\max} \sim a^{-2}$): choose $s = \sigma^2\kappa$ with $s\Lambda_{\max}^2 \approx 1$ for a 50% speed cut; enforce $\sqrt{\int \|F\|^2} \le 0.01\,\text{Area}$ by increasing γ; tie $\Theta \approx \nabla\varphi$ with $\beta \approx \gamma a^2$.

8 Experiments and Reproducibility

Artifacts

We generate the following CSV/PNG files in the working folder:

- `discrete_dirac_pbc_apbc.png`, `dirac_spectrum_pbc_apbc.csv`
- `clock_birth_one_macrocycle.png`, `clock_birth_cover.png`, `clock_birth_signal.csv`

- `binary_holonomy_signflip.png`, `binary_holonomy_test.csv`
- `checksum_mu_sweep.png`

Minimal code (Python)

See the companion script `make_figs_and_data.py`. Run it once to create all artifacts, then compile this `.tex` with `pdflatex` or `xelatex`.

Appendix A: Dirac Spectrum (PBC zero vs APBC gap)

On a discrete circle with N sites and spacing $\Delta\tau = T/N$, eigenvalues are $E(k) = \frac{2}{\Delta\tau}\sin k$, $k = \frac{2\pi}{N}(m + \frac{s}{2})$. Thus $E(0) = 0$ for PBC and $|E| \geq \frac{2}{\Delta\tau}\sin(\pi/N)$ for APBC.

Appendix B: Green Function and Births

For $x_{m+1} - Vx_m + x_{m-1} = s_m$ with $V = 2\cos\phi$, the causal Green function is $G_n = \sin(n\phi)/\sin\phi$. A net-zero s (e.g. $\delta_0 - \delta_{L/2}$) produces a localized birth; an envelope E creates a spiral with pitch set by $\sigma^2\kappa$.

Appendix C: BCH and the Bridge

We record the Strang/BCH expansion $\log U_e = A + B + \frac{1}{12}[B,[B,A]] - \frac{1}{24}[A,[A,B]] + \cdots$ with $A \propto \sigma^2\kappa \mathsf{L}_\alpha^\dagger \mathsf{L}_\alpha$, $B \propto \tau\widehat{K}$.

Appendix D: Pipeline Snippets

```
def invariants_from_stream(y, ell):
    y0, y1, y2 = y[:-2], y[1:-1], y[2:]
    Vhat = (np.vdot(y2, y1) + np.vdot(y0, y1)) / (np.vdot(y1, y1) + 1e-12)
    r_plus = 0.5*(Vhat + np.sqrt(Vhat**2 - 4.0))
    rho, theta = np.abs(r_plus), np.angle(r_plus)
    return Vhat, rho, theta
```